all a coherent vision of the world – with which to resolve the current schizophrenia.

It is not the first time that physics finds itself faced with two highly successful but apparently contradictory theories. The effort to synthesize has in the past been rewarded with great strides forward in our understanding of the world. Newton discovered universal gravity by combining Galileo's parabolas with the ellipses of Kepler. Maxwell found the equations of electromagnetism by combining the theories of electricity and of magnetism. Einstein discovered relativity by way of resolving an apparent conflict between electromagnetism and mechanics. A physicist is only too happy when he finds a conflict of this kind between successful theories: it's an extraordinary opportunity. Can we build a conceptual framework for thinking about the world which is compatible with what we have learnt about it from *both* theories?

Here, in the vanguard, beyond the borders of knowledge, science becomes even more beautiful – incandescent in the forge of nascent ideas, of intuitions, of attempts. Of roads taken and then abandoned, of enthusiasms. In the effort to imagine what has not yet been imagined.

Twenty years ago the fog was dense. Today paths have appeared which have elicited enthusiasm and optimism. There are more than one of these, so it can't be said that the problem has been resolved. The multiplicity generates controversy, but the debate is healthy: until the fog has lifted completely it's good to have criticism and opposing views. One of the principal attempts to solve the problem is a direction of research called 'loop quantum gravity', pursued by a packed squad of researchers working in many countries.

Loop quantum gravity is an endeavour to combine general relativity and quantum mechanics. It is a cautious attempt because it uses only hypotheses already contained within these theories, suitably rewritten to make them compatible. But its consequences are radical: a further profound modification of the way we look at the structure of reality.

The idea is simple. General relativity has taught us that space is not an inert box, but rather something dynamic: a kind of immense, mobile snail-shell in which we are contained – one which can be compressed and twisted. Quantum mechanics, on the other hand, has taught us that every field of this kind is 'made of quanta' and has a fine, granular structure. It immediately follows that physical space is also 'made of quanta'.